# Comparing models with ANOVA (answer key)

Each part has two scenarios with the same tasks. We'll work through scenario **A** together as a class. Then you'll work through scenario **B** in your group.

Every question in this handout compares a **simpler** model to a more **complex** model. Refer back to this box as you go.

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$$F = \frac{(\text{RSS}_{simple} - \text{RSS}_{complex}) \ / \ (p_{complex} - p_{simple})}{\text{RSS}_{complex} \ / \ (n - p_{complex})}$$

- RSS: residual sum of squares, $\sum(y_i - \hat{y}_i)^2$. The variation the model leaves unexplained.
- $p$: number of coefficients the model estimates, including the intercept
- $n$: number of observations
- $df = n - p$: degrees of freedom left over after fitting the model
- **Numerator**: improvement in fit per df spent. **Denominator**: leftover noise per df remaining.

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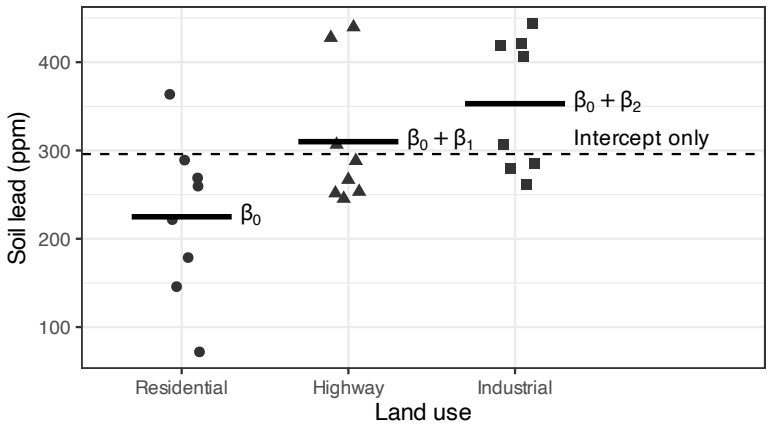
## Part 1: Three levels

For each scenario:

1. **Dummy variables.** Fill in the dummy variable table.
2. **Full model.** Write the equation for the full model.
3. **Draw the means.** Use the coefficients to draw the intercept-only model and each category's mean on the figure.
4. **Degrees of freedom and F.** Fill in $p$ and $df$ for each model, then calculate $F$.
5. **Interpretation.** State $H_0$ and $H_A$. Then interpret the p-value for your $F$ in a sentence.

## 1A. Soil lead (together)

Lead from old gasoline and industrial emissions settles into soil, where children can be exposed. A county health department sampled soil **lead** (ppm) in 8 yards near each of three land uses (**land_use**): residential streets, a highway, and an industrial site.



## Dummy variables

| Land use | Highway | Industrial |
|---|---|---|
| Residential | 0 | 0 |
| Highway | 1 | 0 |
| Industrial | 0 | 1 |

## Full model

$$\text{lead} = \beta_0 + \beta_1 (\text{land use = Highway}) + \beta_2 (\text{land use = Industrial}) + \epsilon$$
$$\epsilon \sim Normal(0, \sigma)$$

## Draw the means

```
coef(lm(lead ~ 1, yards))
```

```
(Intercept)
   295.9625
```

```
coef(lm(lead ~ land_use, yards))
```

```
      (Intercept)    land_useHighway land_useIndustrial
         224.9125            85.0625           128.0875
```

See figure above

## Degrees of freedom and F ($n$ = 24)

| Model | $p$ | $df$ | $RSS$ |
|---|---|---|---|
| `lead ~ 1` | 1 | 23 | 210,538.5 |
| `lead ~ land_use` | 3 | 21 | 142,556.7 |

$$F = \frac{(210,538.5 - 142,556.7) \ / \ (3-1)}{142,556.7 \ / \ (24-3)} = 5.01$$

## Interpretation

State $H_0$ and $H_A$.

$H_0$: $\beta_1 = \beta_2 = 0$

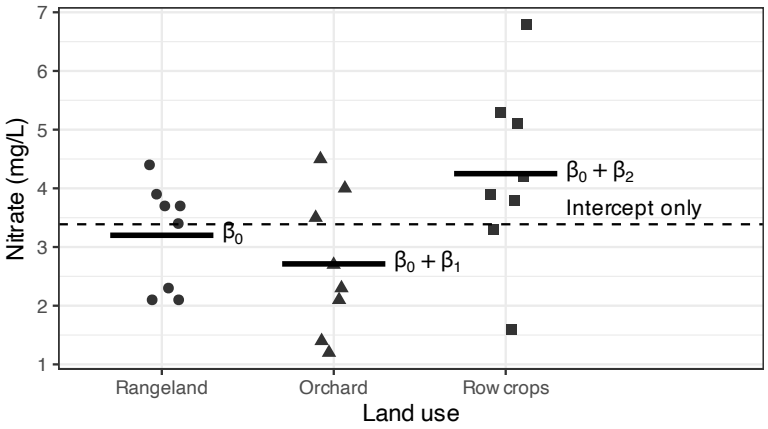
$H_A$: at least one $\beta$ is different than 0.

The p-value for this F is p = 0.017. Interpret it in a sentence.

*If* land use had no relationship with soil lead, we'd see an F at least this large with probability 0.017, so we reject $H_0$ and conclude there is evidence for soil lead varying by land use.

# 1B. Groundwater nitrate (in groups)

Fertilizer applied to crops can leach into groundwater as nitrate, contaminating drinking water wells. A water agency measured **nitrate** (mg/L) in 8 wells under each of three land uses (**land_use**): rangeland, orchards, and row crops.




## Dummy variables

| Land use | Orchard | Row crops |
|---|---|---|
| Rangeland | 0 | 0 |
| Orchard | 1 | 0 |
| Row crops | 0 | 1 |

## Full model

$$\text{nitrate} = \beta_0 + \beta_1 (\text{land use = Orchard}) + \beta_2 (\text{land use = Row crops}) + \epsilon$$
$$\epsilon \sim Normal(0, \sigma)$$

## Draw the means

```
coef(lm(nitrate ~ 1, wells))
```

```
(Intercept)
     3.3875
```

```
coef(lm(nitrate ~ land_use, wells))
```

```
      (Intercept)   land_useOrchard land_useRow crops
           3.2000           -0.4875            1.0500
```

See figure above

## Degrees of freedom and F ($n$ = 24)

| Model | $p$ | $df$ | RSS |
|---|---|---|---|
| `nitrate ~ 1` | 1 | 23 | 42.2 |
| `nitrate ~ land_use` | 3 | 21 | 32.3 |

$$F = \frac{(42.2 - 32.3) \ / \ (3 - 1)}{32.3 \ / \ (24 - 3)} = 3.21$$

## Interpretation

State $H_0$ and $H_A$.

$H_0$: $\beta_1 = \beta_2 = 0$

$H_A$: at least one $\beta$ is different than 0.

The p-value for this F is p = 0.061. Interpret it in a sentence.

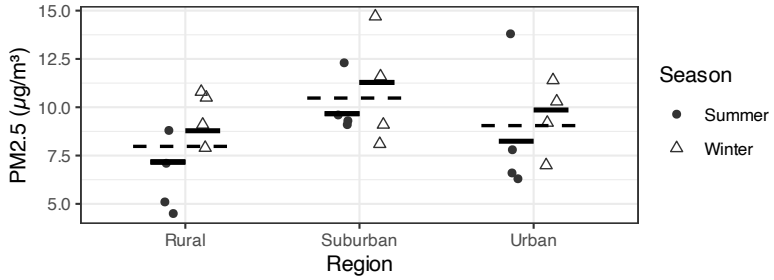
*If* land use had no relationship with nitrate, we'd see an $F$ at least this large with probability 0.061, so we fail to reject $H_0$. There is not enough evidence to conclude that nitrate varies by land use.

# Part 2: Two categorical predictors

Now there are two categorical predictors. Does the second one justify its cost *after* accounting for the first? Same tasks as Part 1, minus the dummy table.

## 2A. Air quality (together)

Fine particulate matter (PM2.5) damages lungs and hearts, and winter wood smoke can trap it near the ground. An air district measured PM2.5 (**pm**, µg/m³) on 4 days for each combination of **region** (rural, suburban, or urban) and **season** (summer or winter). Does season matter after accounting for region?




### Full model

$$\text{pm} = \beta_0 + \beta_1\,(\text{region} = \text{Suburban}) + \beta_2\,(\text{region} = \text{Urban}) + \beta_3\,(\text{season} = \text{Winter}) + \epsilon$$
$$\epsilon \sim Normal(0, \sigma)$$

### Draw the models

```
coef(lm(pm ~ region, air))
```

```
   (Intercept) regionSuburban    regionUrban
         7.975          2.500          1.075
```

```
coef(lm(pm ~ region + season, air))
```

```
   (Intercept) regionSuburban    regionUrban   seasonWinter
      7.166667       2.500000       1.075000       1.616667
```

See figure above

### Degrees of freedom and F ($n$ = 24)

| Model | $p$ | $df$ | RSS |
|---|---|---|---|
| `pm ~ region` | 3 | 21 | 120.4 |
| `pm ~ region + season` | 4 | 20 | 104.7 |

Adding season costs 1 df.

$$F = \frac{(120.4 - 104.7)\ /\ (4 - 3)}{104.7\ /\ (24 - 4)} = 2.99$$

### Interpretation

State $H_0$ and $H_A$.

$H_0$: $\beta_3 = 0$

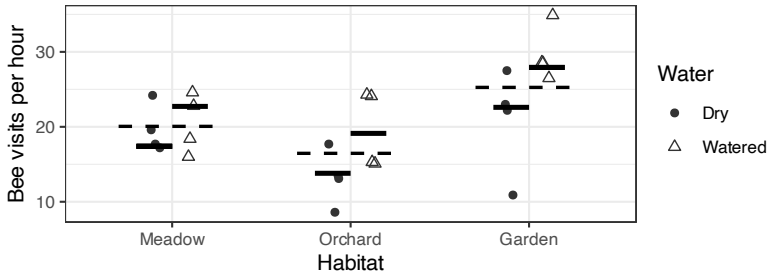
$H_A$: $\beta_3$ is different than 0.

The p-value for this $F$ is p = 0.099. Interpret it in a sentence.

*If* season had no relationship with PM2.5 within each region, we'd see an $F$ at least this large with probability 0.099, so we fail to reject $H_0$. There is not enough evidence to conclude that PM2.5 varies by season after accounting for region.

# 2B. Bees and watering (in groups)

Watered flowers may produce more nectar, attracting more bees. A pollinator team counted bee **visits** per hour to 4 flower patches for each combination of **habitat** (meadow, orchard, or garden) and **water** (dry or watered). Does watering matter after accounting for habitat?




## Full model

$$\text{visits} = \beta_0 + \beta_1\,(\text{habitat = Orchard}) + \beta_2\,(\text{habitat = Garden}) + \beta_3\,(\text{water = Watered}) + \epsilon$$
$$\epsilon \sim Normal(0, \sigma)$$

## Draw the models

```
coef(lm(visits ~ habitat, bees))
```

```
   (Intercept) habitatOrchard  habitatGarden
       20.0625        -3.6000         5.2000
```

```
coef(lm(visits ~ habitat + water, bees))
```

```
   (Intercept) habitatOrchard  habitatGarden   waterWatered
        17.400         -3.600          5.200          5.325
```

See figure above

## Degrees of freedom and F ($n$ = 24)

| Model | $p$ | $df$ | RSS |
|---|---|---|---|
| `visits ~ habitat` | 3 | 21 | 626.7 |
| `visits ~ habitat + water` | 4 | 20 | 456.5 |

Adding watering costs 1 df.

$$F = \frac{(626.7 - 456.5)\ /\ (4-3)}{456.5\ /\ (24-4)} = 7.45$$

## Interpretation

State $H_0$ and $H_A$.

$H_0$: $\beta_3 = 0$

$H_A$: $\beta_3$ is different than 0.

The p-value for this F is p = 0.013. Interpret it in a sentence.

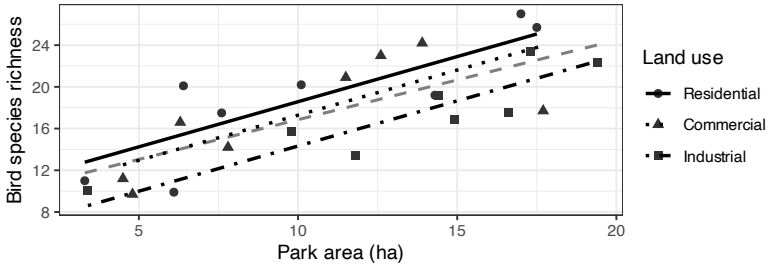
*If* watering had no relationship with bee visits within each habitat, we'd see an F at least this large with probability 0.013, so we reject $H_0$ and conclude there is evidence for bee visits varying with watering after accounting for habitat.

# Part 3: Three levels plus a continuous predictor

Now the second predictor is continuous. Does the categorical predictor justify its cost *after* accounting for the continuous one? Same tasks as Part 2.

# 3A. Birds in city parks (together)

City parks can be refuges for birds, and bigger parks usually hold more species. Does what surrounds a park matter too? A city surveyed bird species **richness** in 8 parks in each of three surrounding land uses (**land_use**): residential, commercial, and industrial. It also recorded each park's **area** (ha).




## Full model

$$\text{richness} = \beta_0 + \beta_1 \text{area} + \beta_2\,(\text{land use = Commercial}) + \beta_3\,(\text{land use = Industrial}) + \epsilon$$
$$\epsilon \sim Normal(0, \sigma)$$

## Draw the models

```
coef(lm(richness ~ area, parks))
```

```
(Intercept)        area
  9.2357591   0.7618654
```

```
coef(lm(richness ~ area + land_use, parks))
```

```
      (Intercept)              area land_useCommercial land_useIndustrial
        9.9183439         0.8657746         -1.2911902         -4.2505121
```

See figure above

## Degrees of freedom and F ($n$ = 24)

| Model | $p$ | $df$ | RSS |
|---|---|---|---|
| `richness ~ area` | 2 | 22 | 257.9 |
| `richness ~ area + land_use` | 4 | 20 | 188.4 |

Adding land use costs 2 df (two dummy variables).

$$F = \frac{(257.9 - 188.4)\ /\ (4-2)}{188.4\ /\ (24-4)} = 3.69$$

## Interpretation

State $H_0$ and $H_A$.

$H_0$: $\beta_2 = \beta_3 = 0$

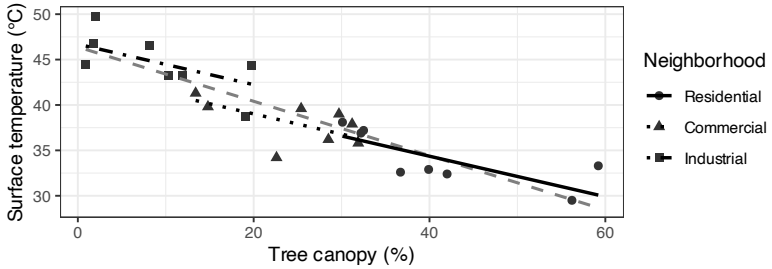
$H_A$: at least one of $\beta_2$ and $\beta_3$ is different than 0.

The p-value for this $F$ is $p = 0.043$. Interpret it in a sentence.

*If* land use had no relationship with richness among parks of the same area, we'd see an $F$ at least this large with probability 0.043, so we reject $H_0$ and conclude there is evidence for bird richness varying by land use after accounting for park area.

# 3B. Urban heat (in groups)

Pavement and rooftops soak up summer sun, while trees shade the ground. A city measured summer surface temperature (**temp**, °C) at 8 sites in each of three **neighborhood** types: residential, commercial, and industrial. It also recorded the tree **canopy** cover (%) around each site. Does neighborhood type matter *after* accounting for canopy?




## Full model

$$\text{temp} = \beta_0 + \beta_1 \text{canopy} + \beta_2\,(\text{neighborhood = Commercial}) + \beta_3\,(\text{neighborhood = Industrial}) + \epsilon$$
$$\epsilon \sim Normal(0, \sigma)$$

## Draw the models

```
coef(lm(temp ~ canopy, sites))
```

```
(Intercept)      canopy
 46.3901866  -0.2991744
```

```
coef(lm(temp ~ canopy + neighborhood, sites))
```

```
          (Intercept)                 canopy neighborhoodCommercial neighborhoodIndustrial
           43.2780196            -0.2230053              0.2024248              3.4194922
```

See figure above

## Degrees of freedom and F ($n$ = 24)

| Model | $p$ | $df$ | RSS |
|---|---|---|---|
| `temp ~ canopy` | 2 | 22 | 122.1 |
| `temp ~ canopy + neighborhood` | 4 | 20 | 96.1 |

Adding neighborhood costs 2 df.

$$\text{F} = \frac{(122.1 - 96.1)\ /\ (4-2)}{96.1\ /\ (24-4)} = 2.70$$

## Interpretation

State $H_0$ and $H_A$.

$H_0$: $\beta_2 = \beta_3 = 0$

$H_A$: at least one of $\beta_2$ and $\beta_3$ is different than 0.

The p-value for this F is p = 0.091. Interpret it in a sentence.

*If* neighborhood type had no relationship with temperature among sites with the same canopy, we'd see an F at least this large with probability 0.091, so we fail to reject $H_0$. There is not enough evidence to

conclude that temperature varies by neighborhood type after accounting for canopy.